

Independent Research Project

Liquidity Provision in Constant-Product AMMs: Impermanent Loss, Fees, and Adverse Selection

A Simulation-Based Analysis of LP Performance
under Alternative Market Regimes

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Abstract

This paper studies the economics of providing liquidity to a constant-product automated market maker (AMM) of the type used in Uniswap V2. The central question is whether arbitrage-driven fee income compensates a liquidity provider (LP) for the structural cost of having pool reserves continuously rebalance against the external market price.

The paper develops a transparent Python simulation framework. It derives the impermanent-loss formula analytically, simulates stochastic price paths under Geometric Brownian Motion (GBM) and a mean-reverting log-Ornstein-Uhlenbeck (log-OU) process, and evaluates LP outcomes through Monte Carlo simulation and market-regime maps in volatility-fee space.

Three extensions enrich the baseline model. First, an exogenous noise-volume channel introduces fee-generating trades that are not directly tied to informational price movements. Second, the analysis compares directional GBM dynamics with mean-reverting log-OU dynamics. Third, it implements a discrete proxy for Loss-Versus-Rebalancing (LVR), following Milionis et al. (2022), in order to isolate the adverse-selection component of AMM rebalancing.

The results identify three central patterns. At typical ETH/USDC parameters, annualized volatility of 60% and a fee rate of 0.30%, the LP underperforms HODL in roughly 70% of simulated paths, with mean underperformance of -3.95% . LP profitability depends not only on volatility, but also on the structure of the price process: mean reversion shifts expected LP outcomes from systematically negative to mildly positive. Finally, the discrete LVR proxy increases substantially faster than fee income as volatility rises, consistent with the quadratic-versus-linear relationship implied by continuous-time theory.

The resulting market-regime maps reproduce the economic logic behind Uniswap V3's tiered fee structure. Low-fee tiers are sustainable at low realized volatility or under strong mean reversion, while volatile pairs require higher fees, significant non-arbitrage volume, or both.

1. Introduction

1.1. Motivation

Decentralized finance has altered market microstructure on public blockchains by replacing traditional limit-order books with automated market makers (AMMs). The dominant AMM family, beginning with Uniswap V2 (Adams et al., 2020), uses a constant-product invariant,

$$x \cdot y = k,$$

to determine asset prices as a function of pool inventory.

This mechanism changes the role of the liquidity provider (LP). An LP deposits two assets into the pool, earns trading fees, and at the same time absorbs the effects generated by arbitrage. The obtained profit is fee income earned passively from third-party trading activity. However, its economic cost is less intuitive.

When the external market price changes, arbitrageurs trade against the AMM until the pool price again matches the market price. In this process, the pool sells the asset that has appreciated and accumulates the asset that has depreciated. Relative to passively holding the original asset portfolio, the LP ends up with a less favourable composition. This mechanism is described as impermanent loss.

The question that motivates this project is whether fee income systematically compensates LPs for this rebalancing cost. The question is empirically and practically important, as billions of dollars sit in AMM pools.

1.2. Research question

The paper addresses one central question and four subsidiary questions.

Central question. Under what market conditions does trading-fee income compensate a liquidity provider for the structural cost of AMM rebalancing?

Additional questions.

1. How does LP profitability depend jointly on volatility and the fee rate?
2. Does exogenous, non-informational trading volume materially change the answer?
3. Does the structure of the price process, namely directional random walk versus mean reversion, change LP outcomes qualitatively rather than merely quantitatively?
4. How does the standard impermanent-loss benchmark compare with Loss-Versus-Rebalancing as a measure of the LP's structural cost?

1.3. Contribution and scope

The project builds a transparent Python simulator and uses it to (a) reproduce the textbook impermanent loss formula, (b) generate stochastic LP outcomes under two price processes, (c) map LP profitability across a two-dimensional regime grid, and (d) implement a discrete-time proxy for LVR that highlights the adverse-selection mechanism in constant-product AMMs.

The scope is deliberately bounded. The model uses a full-range, Uniswap V2-style pool. Fees are treated as accumulated LP income outside the invariant. The model abstracts from gas costs, MEV, concentrated liquidity, dynamic fee policies, and slippage beyond the AMM curve itself.

These restrictions reduce realism in some dimensions, but they make the underlying economic mechanisms transparent. Section 6 discusses the resulting limitations in detail.

2. Theoretical Background

2.1. Constant-product AMM mechanics

A constant-product AMM holds reserves of two assets:

- x : the base asset, for example Ethereum (ETH); and
- y : the quote asset, for example USD Coin (USDC).

The defining invariant is:

$$x \cdot y = k,$$

where k is the pool's liquidity constant.

The marginal pool price of the base asset, measured in quote-currency units, is:

$$P = \frac{y}{x}.$$

When the external market price P' differs from the AMM price, arbitrageurs trade against the pool until the two prices align. After equilibration, the pool must satisfy both:

$$x' y' = k$$

and

$$P' = \frac{y'}{x'}.$$

Solving this system yields:

$$x' = \sqrt{\frac{k}{P'}}, \quad y' = \sqrt{kP'}.$$

These reserve levels are not simulation results. They follow directly from the constant-product invariant. Conditional on k , the pool composition is a deterministic function of the external price.

An important property of this mechanism is that the AMM has no direct access to external market prices. The pool is a deterministic smart contract whose state — the reserves x and y — can only be updated through transactions submitted to it. There is no mechanism by which the pool can query a centralized exchange or any other off-chain price source. The implication is that the pool's internal price reflects only its own trading history, not the current external market. External price information enters the pool exclusively through arbitrage activity: when the external price diverges from the AMM price, arbitrageurs trade against the pool until alignment is restored. This makes arbitrage the sole channel of price discovery for the AMM, and the resulting structural cost to the LP is the subject of Sections 2.2 to 2.4.

2.2. Arbitrage and structural inventory rebalancing

The equilibrium expressions imply two immediate properties.

First, the pool's base-asset reserve decreases as the external price rises:

$$x' = \sqrt{\frac{k}{P'}}.$$

When ETH appreciates, the pool ends up holding less ETH, as it sold ETH to the arbitrageur.

Second, the quote-asset reserve increases as the external price rises:

$$y' = \sqrt{kP'}.$$

The pool thus holds more USDC and less ETH precisely when ETH has become more valuable. The mechanism operates symmetrically when the price falls.

In other words, arbitrage has two effects at once. It keeps the AMM aligned with the external market, but it imposes a structural cost on the LP. Meanwhile, the LP stands on the opposite side of every arbitrage trade and systematically sells the rising asset while accumulating the falling asset.

The compensation for this role is the trading fee. A fraction ϕ of every trade through the pool accrues to the LP. In actual Uniswap implementations, fees remain inside the pool and increase k . The simulator at hand instead records fees as a separate income stream outside the invariant. This simplification improves analytical transparency and is revisited in Section 6.

2.3. The impermanent loss formula

Let the initial pool reserves be (x_0, y_0) , with initial price:

$$P_0 = \frac{y_0}{x_0}.$$

Define the price ratio as:

$$r = \frac{P_T}{P_0}.$$

Ignoring fees, the value of the LP position at horizon T is:

$$V_{LP}(P_T) = x_T P_T + y_T = 2\sqrt{kP_T} = 2y_0\sqrt{r}.$$

The value of passively holding the initial asset quantities is:

$$V_{HODL}(P_T) = x_0 P_T + y_0 = y_0(1 + r).$$

Impermanent loss is the relative gap between the two values:

$$IL(r) = \frac{V_{LP}}{V_{HODL}} - 1 = \frac{2\sqrt{r}}{1 + r} - 1.$$

The formula has three properties that determine the logic of the rest of the analysis.

First, impermanent loss depends only on the final price ratio r . It does not hinge on the price path, the initial reserve size, or the absolute price level. Any 50/50 constant-product LP position with the same terminal price ratio incurs the same percentage impermanent loss.

Second, the formula is symmetric in r and $1/r$. A price doubling, $r = 2$, and a price halving, $r = 0.5$, both generate impermanent loss of approximately -5.72% .

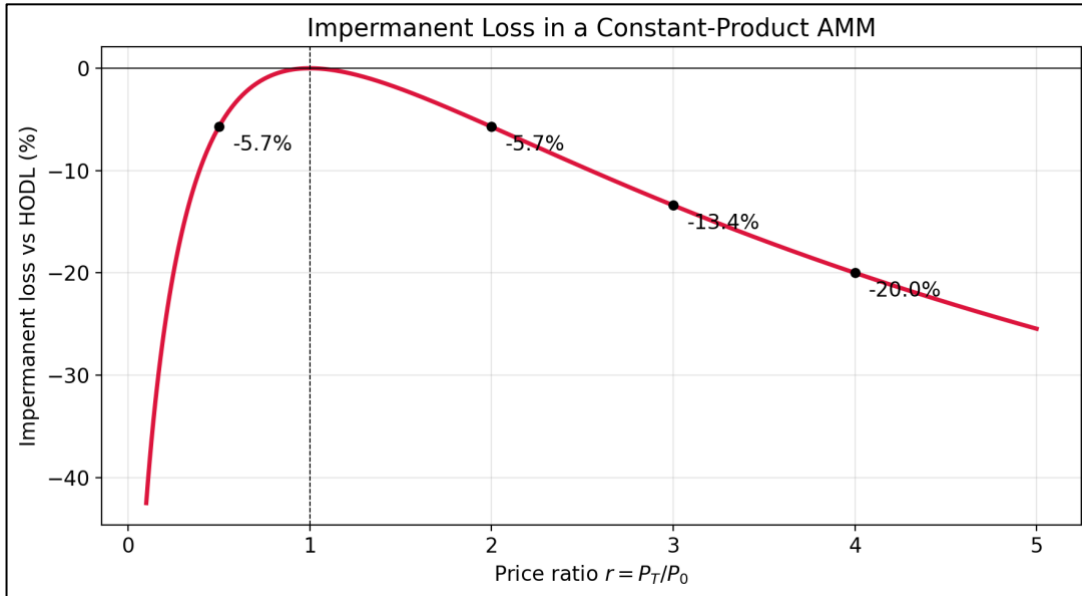
Third, impermanent loss is non-positive and equals zero only when $r = 1$. The arithmetic-mean-geometric-mean inequality implies:

$$2\sqrt{r} \leq 1 + r,$$

and therefore:

$$IL(r) \leq 0.$$

Figure 1: *Impermanent loss as a function of price ratio*



2.4. Fees and the asymmetric scaling problem

The structural cost of liquidity provision is driven by price movements, while fee income is driven by trading volume.

In a constant-product AMM, the volume required to move the price by a given amount is linked to the resulting change in pool reserves. In the pure-arbitrage model used here, fee income therefore grows approximately linearly with volatility σ .

Impermanent loss behaves differently. As Section 4 demonstrates, expected LP underperformance grows convexly with volatility and is approximately quadratic in σ .

This difference creates the central asymmetry of the model:

fee income is proportional to σ ,

while

structural LP loss is proportional to σ^2 .

For any fixed fee rate ϕ , a volatility threshold thus exists beyond which LP-with-fees underperforms HODL on average. The relevant question is where this threshold lies under economically plausible parameter values.

3. Methodology

3.1. Price processes

The model uses two stochastic processes for the base-asset price.

Geometric Brownian Motion (GBM)

Geometric Brownian Motion provides the benchmark for a positive asset price with directional uncertainty:

$$\frac{dP}{P} = \mu dt + \sigma dW.$$

With daily time steps, the discretised process is:

$$P_{t+\Delta t} = P_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z \right], \quad Z \sim \mathcal{N}(0,1).$$

The term $-\frac{1}{2} \sigma^2$ is the so-called Itô correction. It ensures consistency between the discrete implementation and the continuous-time process.

Log-Ornstein-Uhlenbeck process (log-OU)

The second price process is a mean-reverting log-Ornstein-Uhlenbeck specification:

$$d(\log P) = \kappa(\log \mu_\infty - \log P)dt + \sigma dW.$$

The process represents asset pairs with a long-run price anchor, such as stablecoin pairs or wrapped-token pairs.

The reversion parameter κ determines the speed at which price deviations decay. Its half-life is:

$$t_{1/2} = \frac{\ln(2)}{\kappa}.$$

The process is specified in log-price space in order to preserve strict positivity. The volatility parameter σ is measured in the same annualized relative units as in GBM, which makes the two specifications directly comparable.

The qualitative difference is substantial. While GBM permits persistent deviations from the initial level, the log-OU process repeatedly pulls price back toward the long-run anchor. The GBM is the appropriate model for assets without a structural price anchor, such as ETH/USDC, BTC/USDC, or most fiat-quoted crypto pairs. The log-OU process reflects well assets with an economic peg, such as USDC/USDT (stablecoin pairs), ETH/stETH (liquid staking derivatives), or wBTC/BTC (wrapped tokens).

Figure 2a: Five GBM price paths

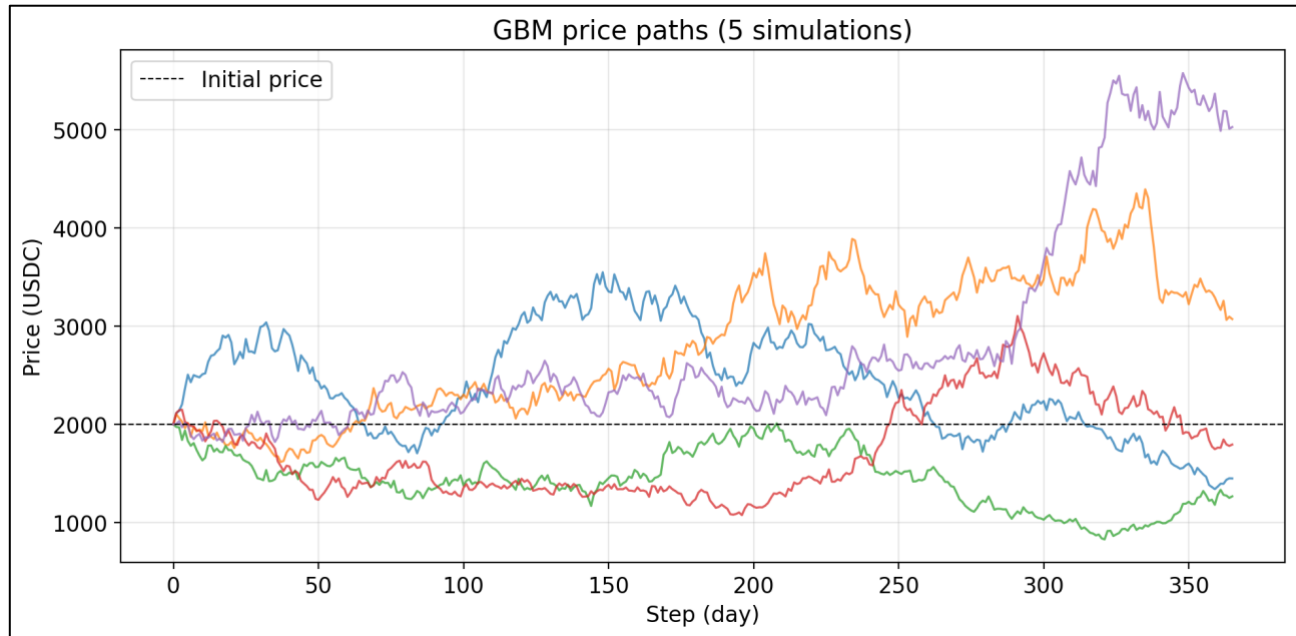
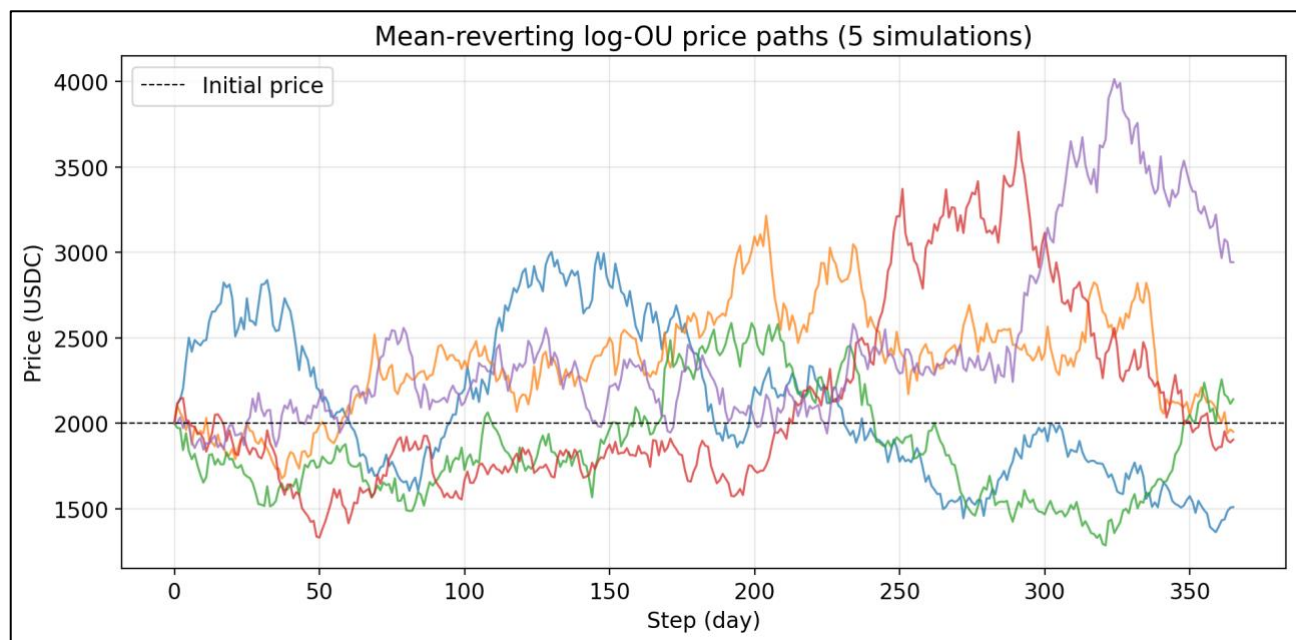


Figure 2b: Five mean-reverting log-OU price paths



3.2. LP simulation

For every simulated price path, the model advances one time step at a time.

At each step, the following sequence applies.

Firstly, the pool reserves rebalance to the values implied by the new external price:

$$x' = \sqrt{\frac{k}{P'}}, \quad y' = \sqrt{kP'}.$$

This represents the post-arbitrage pool state.

Secondly, the absolute change in the quote reserve (the current amount of the quote asset),

$$|\Delta y|,$$

serves as a transparent proxy for arbitrage volume in quote-currency units.

Thirdly, the corresponding arbitrage fee income is:

$$\phi \cdot |\Delta y|.$$

Fourthly, the optional noise-volume channel adds:

$$\eta \cdot \text{TVL} \cdot \Delta t$$

of quote-currency trading volume per time step. This volume earns fees at the same rate as arbitrage volume but does not change prices or reserves in the simplified framework.

Finally, the model records a discrete Loss-Versus-Rebalancing proxy:

$$\Delta x \cdot P' - \Delta y.$$

The expression captures the difference between the fair-market value of the base asset transferred by the pool (in the case at hand, ETH) and the quote asset received in exchange (USDC).

Each simulation therefore produces four terminal values:

1. the HODL value;
2. the LP value before fees;
3. the LP value after fees;
4. the accumulated discrete LVR proxy.

3.3. Monte Carlo design

Although a single simulated path can provide intuition, it does not reveal the distribution of outcomes. For this reason, the analysis uses Monte Carlo simulation with vectorised batch path generation.

For each parameter combination, n independent paths generate n LP outcomes. The paper reports:

- the mean LP-minus-HODL outcome;
- the median outcome;
- the LP win rate;
- the 5th percentile;
- the 95th percentile.

The LP **win rate** is defined as the share of simulations in which LP-with-fees exceeds HODL.

For the regime maps, each (σ, ϕ) cell uses a separate seed offset. This prevents neighbouring cells from relying on the same random draws and avoids artificial smoothness in the visual output.

3.4. Implementation

The price generators are vectorized across simulations. Depending on the selected run mode, a 12×7 regime map evaluates either 8,400 full-year simulations in quick mode or 25,200 in full mode. The full source code, generated figures, and reproducibility instructions are available in the accompanying GitHub repository.

Table 1 summarizes the baseline parameters.

Table 1: Baseline parameters

Parameter	Value
Initial ETH price P_0	2,000 USDC
Initial reserves (x_0, y_0)	1 ETH, 2,000 USDC
Initial portfolio value	4,000 USDC
Horizon T	1 year
Time steps	365 daily
Baseline volatility σ	60% annualised
Baseline fee rate ϕ	0.30%
GBM drift μ	0%
OU long-run price μ_∞	2,000 USDC
OU reversion speed κ	4.0 per year (half-life ≈ 63 days)

4. Base Results

4.1. Single-path inspection

Figure 3 shows one simulated GBM path using seed 42. The price falls from 2,000 to a trough near 1,200, recovers to approximately 2,100, and ends at 1,872.

The lower panel compares three portfolio values:

- HODL;
- LP without fees;
- LP with fees.

The price does not finish far from its initial level. The terminal price ratio is:

$$r = 0.94,$$

which implies static impermanent loss of approximately -0.05% . The three portfolio lines therefore remain close on the displayed scale.

Figure 4 separates the two mechanisms hidden in the aggregated portfolio values.

The red line shows LP-without-fees minus HODL. By construction, it represents instantaneous impermanent loss measured in USDC.

The green line shows LP-with-fees minus HODL.

The shaded area between the two lines represents cumulative fee income.

Figure 3: Price path and the three strategies

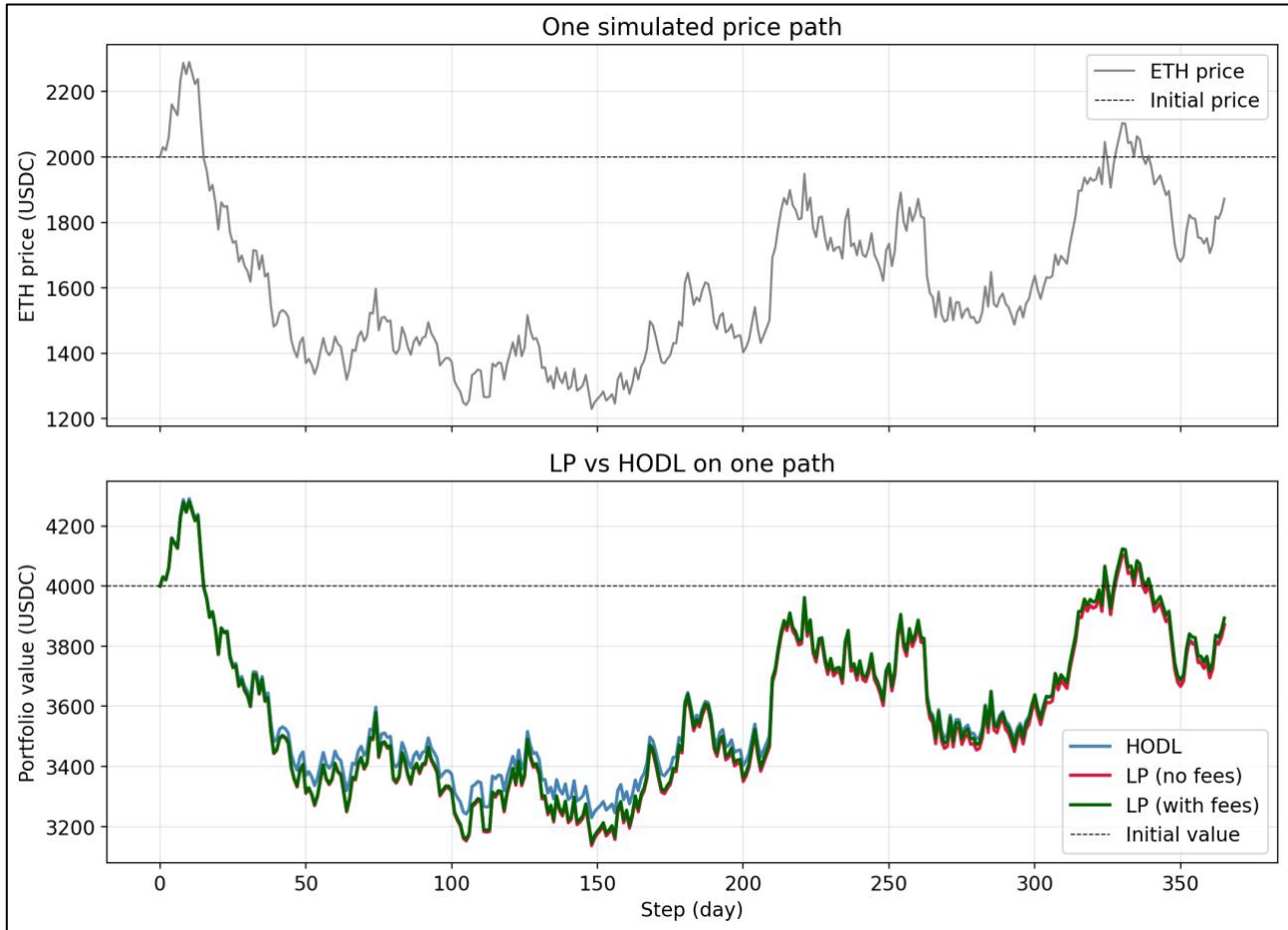


Figure 4: Impermanent Loss versus accumulated fees on one path

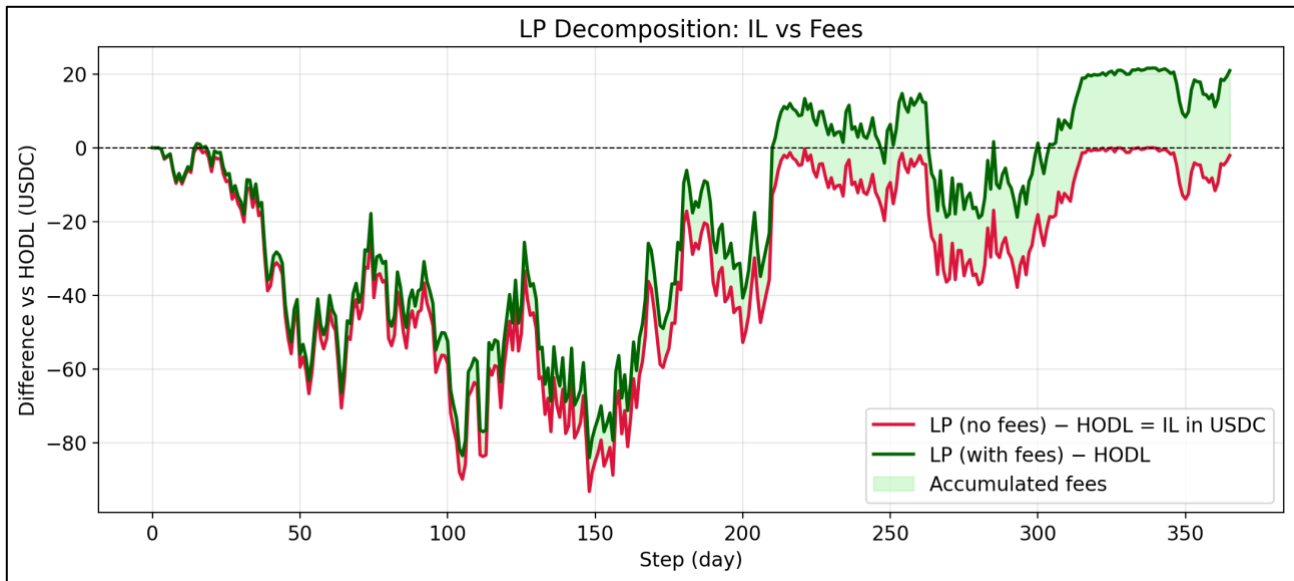


Figure 4 separates the two mechanisms hidden in the aggregated portfolio values.

The red line shows LP-without-fees minus HODL. By construction, it represents instantaneous impermanent loss measured in USDC. The green line shows LP-with-fees minus HODL. The shaded area between the two lines represents cumulative fee income. The two lines remain close near the beginning and end of the path because the price is close to P_0 , so impermanent loss is small. The gap widens around day 150, when the price moves further away from its initial level and the inventory effect becomes economically meaningful.

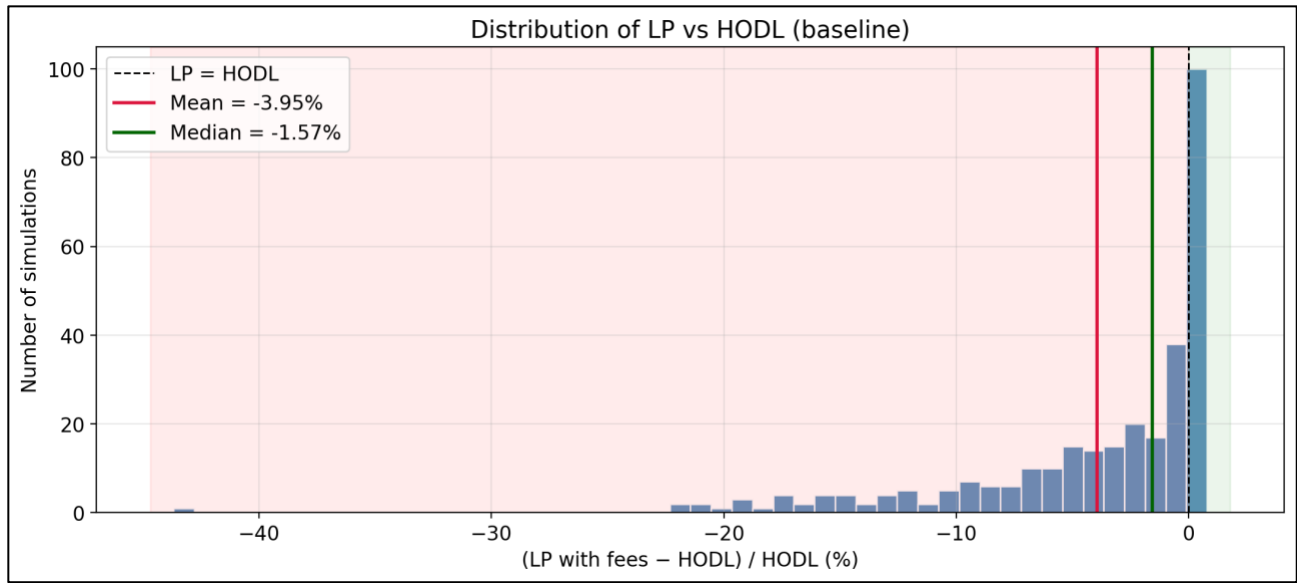
Fees behave differently. They accumulate monotonically and never decline. By the end of the year, accumulated fees reach approximately 23 USDC. They almost, but not fully, offset the impermanent-loss gap. The LP-with-fees position therefore finishes close to HODL.

The single-path analysis captures the core economic tension. Fees accumulate continuously, while the inventory cost becomes significant when the price moves materially away from its starting point.

4.2. Distributional outcomes

Figure 5 shows the distribution of LP-versus-HODL outcomes across 300 independent simulations at baseline parameters.

Figure 5: Histogram of LP outcomes at baseline



The distribution has four important features.

Firstly, it is strongly left-skewed. The left tail extends to roughly -45% , while the right tail barely reaches $+2\%$.

Secondly, the mean outcome of -3.95% lies materially below the median outcome of -1.57% . Rare but severe price moves pull down the mean without affecting the median to the same extent.

Thirdly, the modal outcome lies near zero. Most paths end relatively close to the initial price. In these cases, impermanent loss remains limited and accumulated fees broadly compensate for it.

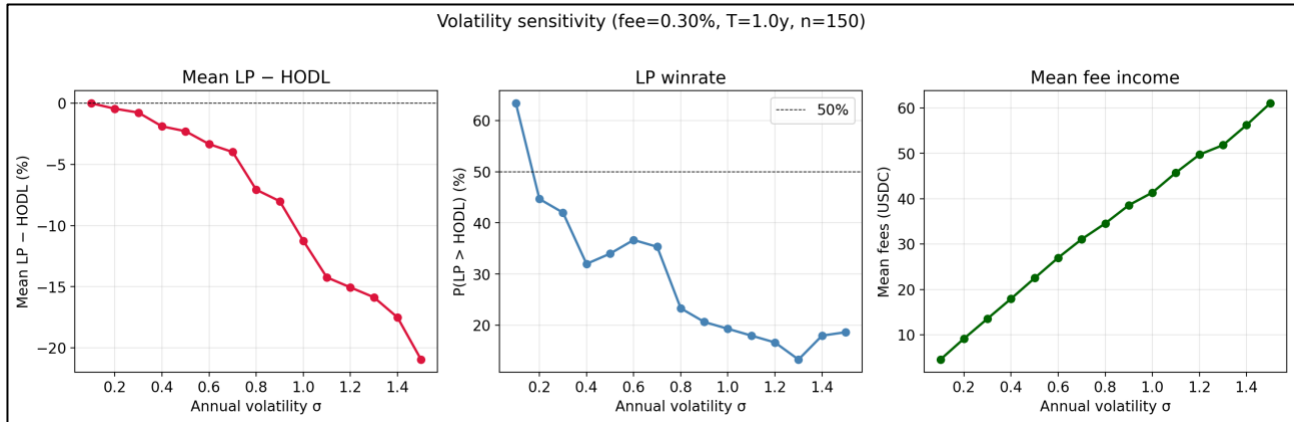
Fourthly, the LP win rate is approximately 31%. The LP therefore outperforms HODL in fewer than one third of simulated paths.

The asymmetry is structural. Impermanent loss cannot generate positive inventory outperformance because it is bounded above by zero. It can, however, become increasingly negative as the terminal price diverges from the initial price.

4.3. Volatility sensitivity

Figure 6 shows how mean LP-minus-HODL performance, LP win rate, and mean fee income change as annualised volatility rises from 10% to 150%, while the fee rate remains fixed at 0.30%.

Figure 6: Three-panel volatility sweep



The left panel shows the central asymmetry of the model. Mean LP-minus-HODL falls from close to zero at $\sigma = 0.1$ to approximately -21% at $\sigma = 1.5$. The pattern is clearly convex.

The middle panel shows a monotonic decline in the LP win rate. It falls from roughly 63% at $\sigma = 0.1$ to below 20% at $\sigma = 1.0$. The win rate crosses the 50% threshold between:

$$\sigma \approx 0.10 \text{ and } \sigma \approx 0.20.$$

Above this level, LP loses to HODL more often than it wins at the baseline fee rate.

The right panel shows that mean fee income rises approximately linearly with volatility. Fees increase from roughly 5 USDC at $\sigma = 0.1$ to roughly 60 USDC at $\sigma = 1.5$.

The implication is:

fee income rises approximately linearly,

while

LP loss rises convexly.

Linear income cannot indefinitely compensate for convex structural loss.

4.4. Market-regime map

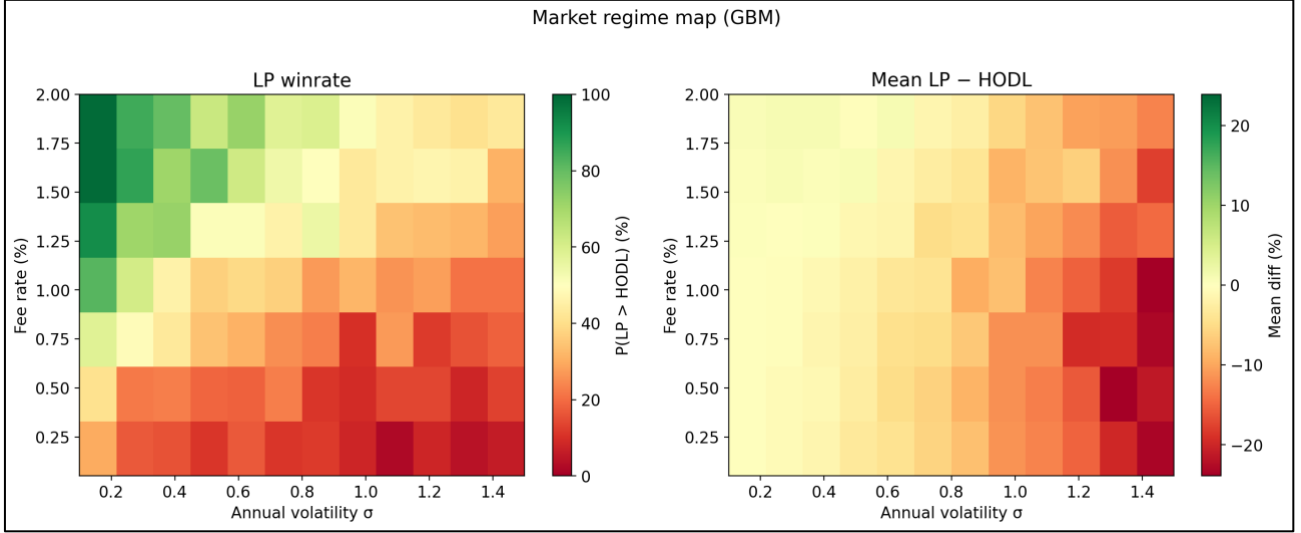
Figure 7 extends the volatility sweep by varying volatility and fee rates jointly.

Each cell reports:

- LP win rate in the left panel;
- mean LP-minus-HODL outcome in the right panel.

Every cell is based on a 300-simulation Monte Carlo experiment.

Figure 7: Market-regime map under GBM



As shown in Figure 7, the favorable LP region (deep green) is concentrated in the upper-left corner: low volatility combined with high fees. The unfavorable region (deep red) lies in the lower-right corner: high volatility combined with low fees. The boundary slopes upward because more volatile pairs require higher fee rates to make LP competitive with HODL.

The mapping to real-world AMM design is qualitative rather than direct. Uniswap V3 (Adams et al., 2021) introduced multiple fee tiers, reflecting the idea that pairs with different volatility and trading-volume profiles may require different fee schedules. Figure 7 is qualitatively consistent with this intuition: holding the remaining model assumptions fixed, more volatile pairs require higher fees to make liquidity provision competitive with HODL. The figure does not estimate an optimal fee tier or prescribe a fee for any actual pool.

The baseline ETH/USDC point,

$$\sigma = 0.6, \quad \phi = 0.3\%,$$

lies firmly in the unfavorable (red) region, with a win rate near 31%, while a mean LP-HODL of approximately -4% . Existing empirical work on selected Uniswap V3 pools finds that aggregate impermanent loss can exceed earned fees (Loesch et al., 2021). The present ETH/USDC-style result is a stylised simulation benchmark rather than an empirical estimate of realised pool performance.

5. Extensions

5.1. Noise-trader volume

The baseline model attributes all trading volume to arbitrageurs, whose activity is directly tied to price movements. In reality, a substantial share of pool volume comes from end-users whose swaps are uncorrelated with informational price changes. They might be just paying for goods or services in stablecoins, on/off-ramping, providing liquidity for unrelated DeFi operations. This noise volume earns fees for LPs without contributing to adverse selection.

The extension introduces the parameter η , defined as the daily fraction of pool TVL traded as noise volume. For ETH/USDC-style pools on Uniswap, the model considers noise-volume scenarios:

$$\eta = 0.3 \text{ to } 1.0,$$

or 30% to 100% of pool TVL per day. These values are used as scenario inputs for the analysis.

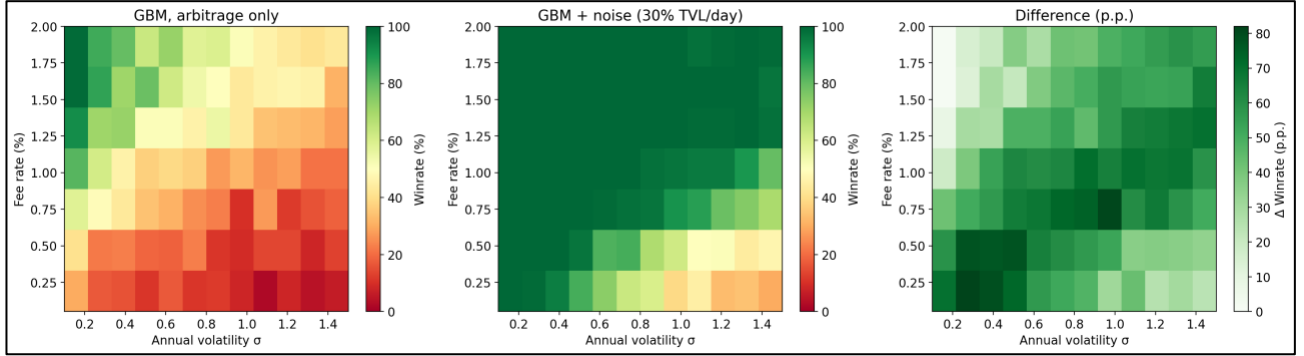
The model adds:

$$\eta \cdot \text{TVL} \cdot \Delta t$$

of fee-generating volume at each time step. Prices and reserves remain unchanged. Figure 8 compares the pure-arbitrage regime map with the same map at:

$$\eta = 0.30.$$

Figure 8: Comparison without/with noise volume and resulting win-rate gain



Two observations follow. First, the favorable LP region expands considerably. Under the model’s noise-volume assumption, the upper half of the map becomes green in most cells, including high-volatility regions with fee rates of 1% or above. Parts of the middle and lower-right regions also become viable. The win-rate gain (right panel) reaches roughly 70–80 percentage points in some cells.

Second, the gain is largest where arbitrage volume was small, namely the left edge, where volatility is low. There, arbitrage by itself generates almost no fee income; noise fills the gap. At high σ , arbitrage already generates substantial fees, and the relative contribution of noise is smaller.

The economic implication is that the empirical viability of low-fee stablecoin pools fundamentally depends on non-arbitrage trading volume. Without it, even pools with volatility less than 10% and fees of 5 basis points would likely lose to HODL on average; with the model’s assumed noise levels, they are robustly profitable.

5.2. Mean-reverting prices

Figure 9 compares LP outcome distributions under GBM and log-OU dynamics at the same baseline parameters. The shift is substantial.

Under GBM:

$$\text{mean LP-minus-HODL} = -3.95\%,$$

and

$$\text{LP win rate} = 31\%.$$

Under log-OU:

$$\text{mean LP-minus-HODL} = +0.12\%,$$

and

LP win rate = 72%.

The two distributions therefore represent different economic regimes rather than small variations around the same baseline.

Figure 9: Distribution comparison

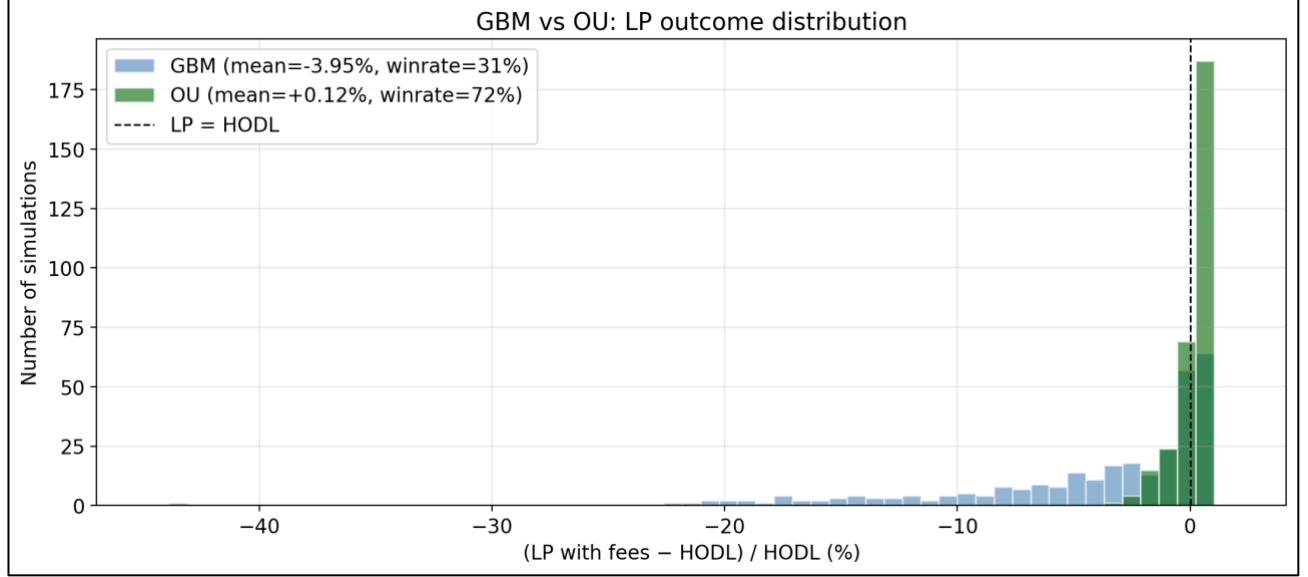


Figure 10: Regime maps for GBM and log-OU

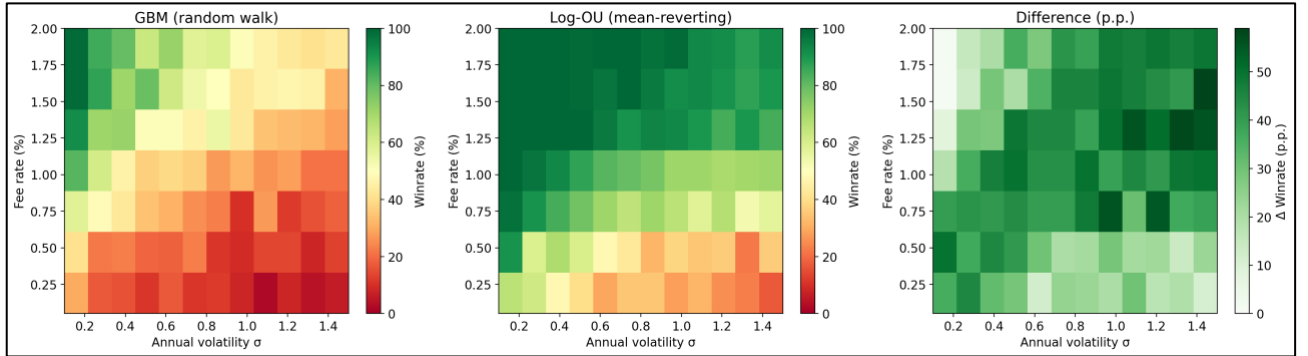


Figure 10 extends the comparison to the full volatility-fee map.

Under log-OU dynamics, the favorable LP region covers most of the map. Even at high volatility,

$$\sigma = 1.5,$$

a fee rate of 1% or above produces LP win rates above 80%.

The gain from mean reversion is especially large in the high-volatility region. There, switching from GBM to log-OU raises the LP win rate by approximately 40 to 55 percentage points. This is the largest qualitative shift induced by any extension in this project. The mechanism follows directly from the definition of impermanent loss, which depends on the terminal price ratio:

$$r = \frac{P_T}{P_0}.$$

Under GBM, the price can end far from its initial level. This produces large terminal impermanent loss.

Under log-OU, price deviations repeatedly revert toward the long-run anchor. The price may move substantially during the year, generating arbitrage volume and fees, but the terminal price is more likely to remain close to the initial level. Terminal impermanent loss thus remains limited.

Hence, mean reversion creates a particularly favorable LP environment: volatility generates fee income without producing equally large terminal IL.

This result helps explain why stablecoin pools and liquid-staking-derivative pools can operate at low fee tiers. Their relative prices are constrained by economic equivalence or stabilisation mechanisms and therefore behave more like mean-reverting processes than directional asset pairs.

5.3. Loss-Versus-Rebalancing

Impermanent loss compares the LP position with passive HODL. HODL, however, remains exposed to price risk and does not maintain a balanced portfolio over time.

Loss-Versus-Rebalancing (LVR), introduced by Milionis, Moallemi, Roughgarden, and Zhang (2022), uses a stricter benchmark. Namely, it compares the AMM LP with a portfolio that continuously maintains a 50/50 value allocation but rebalances at the fair external price rather than at the AMM's stale internal price.

The difference between the two, namely the value lost because the AMM rebalances against an out-of-date price, is called Loss-Versus-Rebalancing.

For a constant-product AMM, continuous-time theory gives the instantaneous LVR rate:

$$\text{LVR}_t = \frac{\sigma^2 P_t x_t}{8}.$$

The expression is explicitly quadratic in volatility.

The present analysis implements the following discrete proxy:

$$\text{LVR-step}_t = \Delta x_t \cdot P_{t+1} - \Delta y_t.$$

The expression measures the difference between:

- the fair-price value of the base asset transferred by the pool;
- the quote asset actually received by the pool.

Summed over a path, the proxy gives a measure of cumulative arbitrage-driven value leakage.

Figure 11 shows the proxy on the same path used in Section 4.1, together with cumulative fees.

By the end of the year, cumulative LVR reaches 144.6 USDC, while cumulative fees amount to 23.0 USDC.

Fees therefore recover approximately:

$$\frac{23.0}{144.6} \approx 16\%$$

of the structural leakage.

Figure 11: Cumulative LVR proxy versus cumulative fees

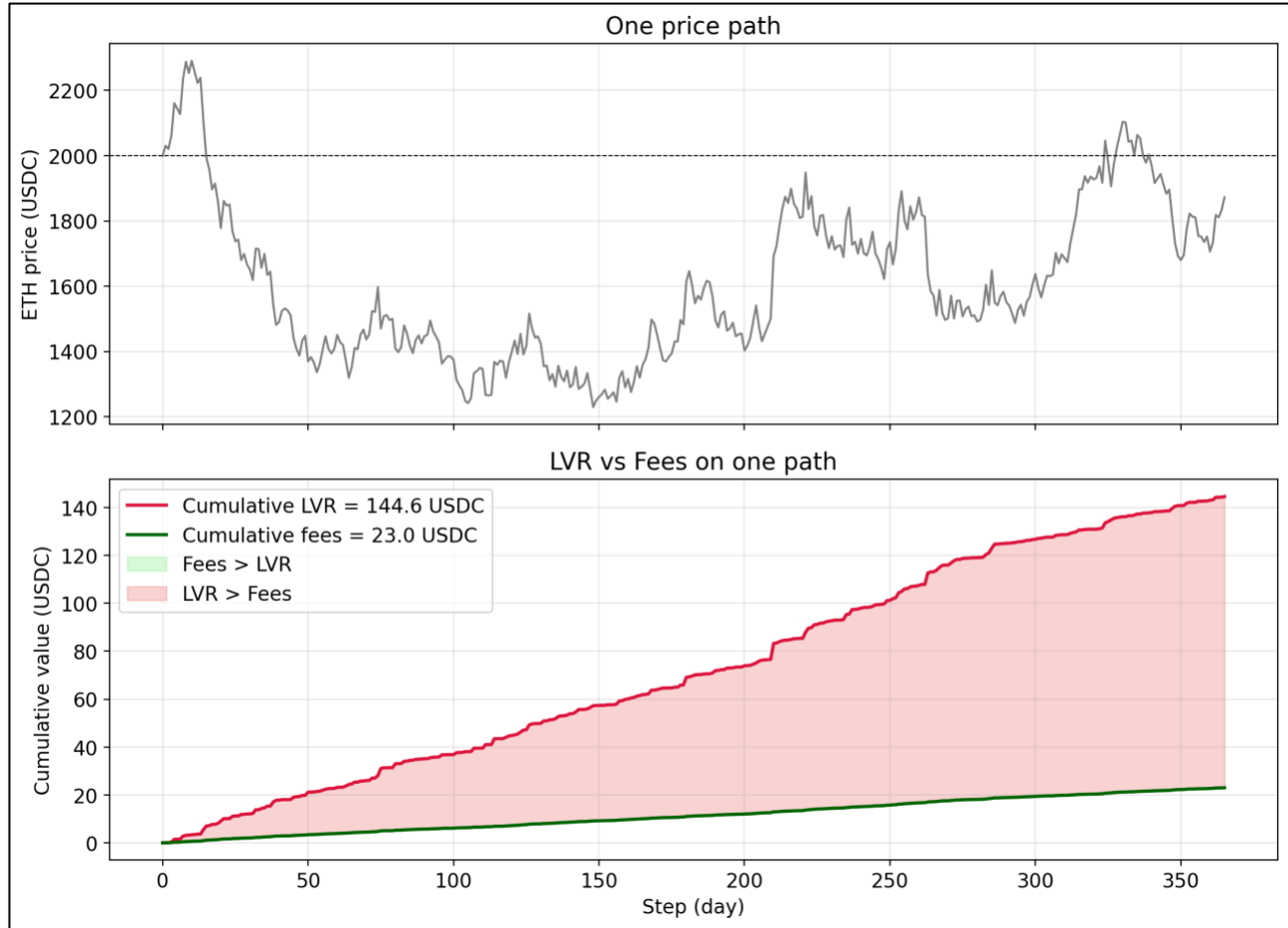
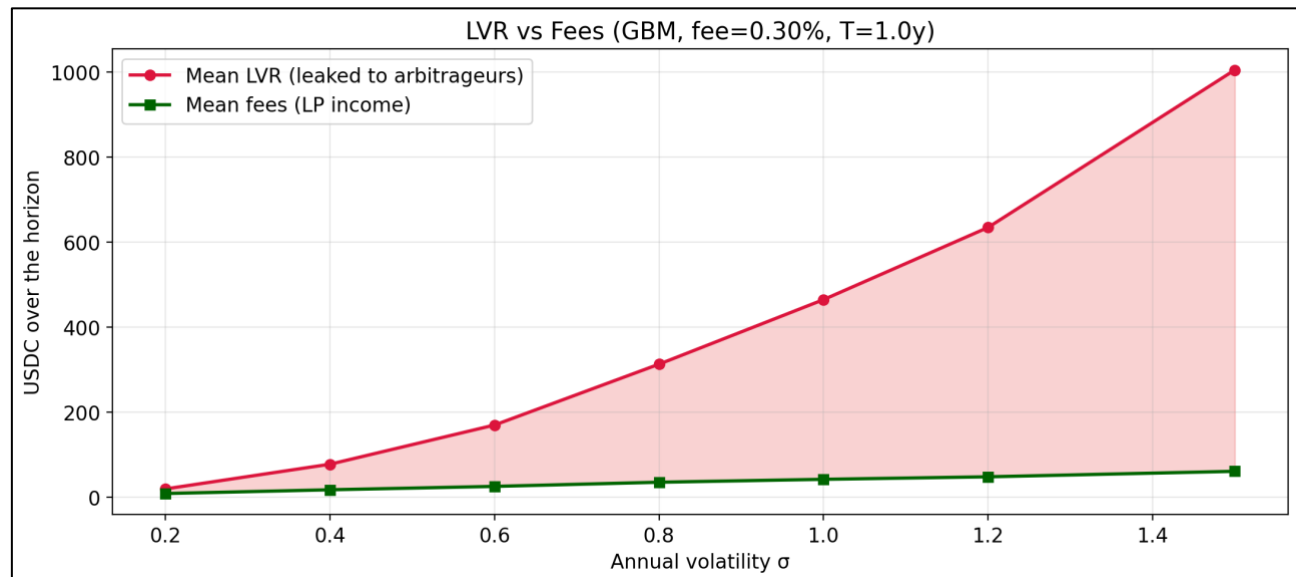


Figure 12: LVR and fees as functions of volatility



The LVR-versus-fees gap is monotonic over time and considerably larger than the IL-versus-fees gap shown in Figure 4. This is expected because LVR uses a stricter benchmark than HODL.

Figure 12 extends the analysis across volatility levels. Both LVR and fee income are averaged over 150 simulations for each value of σ . The LVR curve rises convexly, from roughly 20 USDC at $\sigma = 0.2$ to approximately 1,000 USDC at $\sigma = 1.5$. The fee curve rises much more slowly, from roughly 8 USDC to 60 USDC across the same range. The fee-to-LVR ratio thus declines sharply:

- approximately 50% at $\sigma = 0.2$;
- approximately 6% at $\sigma = 1.5$.

Two conclusions follow.

Firstly, the fee-to-LVR ratio is a stricter measure than the LP-versus-HODL win rate. At baseline volatility,

$$\sigma = 0.6,$$

the LP win rate is approximately 31%. This may suggest moderate underperformance. The fee-to-LVR ratio, however, is only around 17%, indicating that more than 80% of structural value leakage remains uncompensated.

Secondly, the quadratic-versus-linear scaling problem appears again. LVR rises approximately quadratically with volatility, while fees rise approximately linearly. No fixed fee rate can indefinitely compensate for a quadratic increase in adverse-selection cost.

The discrete proxy remains deliberately transparent rather than exact. Continuous-time LVR requires Itô calculus and specific assumptions about the underlying process. The proxy does not reproduce the exact continuous-time value, but it preserves the relevant economic mechanism and volatility scaling.

6. Discussion

6.1. Synthesis

Six findings emerge from the analysis.

Firstly, the impermanent-loss formula is symmetric and depends only on the terminal price ratio. A price doubling and a price halving generate identical percentage losses relative to HODL, independently of path or position size.

Secondly, under baseline ETH/USDC-style parameters,

$$\sigma = 60\%, \quad \phi = 0.30\%,$$

the LP underperforms HODL on average. Mean underperformance is approximately 4%, and the LP win rate is approximately 31%.

Thirdly, the volatility-fee regime map reproduces the economic logic of Uniswap V3 fee tiers. Low-fee pools remain sustainable primarily at low volatility. High-volatility pairs require higher fees in order to make LP competitive with HODL on average.

Fourthly, non-arbitrage volume substantially expands the favorable LP region. At noise-volume levels around 30% of TVL per day, the upper half of the regime map becomes favourable in many cases. This helps explain why low-fee stablecoin pools can work in practice.

Fifthly, price-process structure matters at least as much as volatility itself. Switching from GBM to a mean-reverting log-OU process shifts the baseline LP win rate from 31% to 72% and moves mean LP-minus-HODL from approximately -4% to around zero.

Finally, the LVR proxy shows that fees recover only a limited fraction of structural leakage. At the baseline ETH/USDC point, fees recover approximately 16% of LVR. At $\sigma = 1.5$, they recover less than 7%.

The listed results are mutually consistent. The impermanent-loss formula provides the static structure of the problem. The GBM regime map extends this structure dynamically. Noise volume and mean reversion describe two distinct mechanisms that can make LP positions viable. LVR then provides the most direct measure of the value lost through stale-price AMM rebalancing.

6.2. Limitations

The model relies on several simplifying assumptions.

Fees are treated as income outside the invariant k . In Uniswap V2, fees remain inside the pool and increase k over time, generating a compounding effect. This affects precise numerical magnitudes, although not the core economic trade-off.

The noise-volume extension generates fees without changing the pool price or reserves. In reality, every swap moves the pool, and a later arbitrage trade partially reverses this movement while generating additional fees. The simplification thus slightly understates the fee contribution of noise volume.

The price processes are stylized. GBM does not capture volatility clustering, jumps, fat tails, or regime changes. The log-OU process provides a useful comparative regime, but does not represent a literal stablecoin model because stablecoin prices are governed by explicit stabilisation mechanisms.

The model uses a full-range Uniswap V2-style pool. Uniswap V3 introduces concentrated liquidity. LPs can earn substantially higher fees inside a chosen range, but may earn reduced or zero fees once the price leaves that range. This creates a different risk profile.

Several practical frictions are excluded: gas costs; MEV and sandwich attacks; dynamic fees; liquidity migration; strategic LP behavior; slippage beyond the constant-product curve.

Finally, the LVR calculation is a discrete proxy. Continuous-time LVR is exact only under specific stochastic-calculus assumptions. The proxy preserves the central mechanism and scaling pattern, but its numerical level depends on the time-step size.

6.3. Future work

Four extensions would be natural next steps.

1. **Empirical calibration.** Estimating volatility and noise-volume parameters from historical Uniswap ETH/USDC data would transform the regime maps from analytical illustrations into empirically grounded prediction tools.
2. **Uniswap V3 concentrated liquidity.** Adding range-bound LP positions would extend the model to the dominant production AMM design and allow direct comparison between passive and actively managed liquidity provision.
3. **Stochastic volatility and jumps.** Crypto-asset prices exhibit volatility clustering and discontinuous jumps. Heston-style stochastic volatility or Merton-style jump-diffusion processes would create a more realistic price environment.

4. **MEV-aware fee accounting.** A meaningful share of arbitrage value in decentralised markets is affected by searcher competition, transaction ordering, and validator extraction. Incorporating these mechanisms would refine the relationship between fee income and LVR.

7. Conclusion

The project asked when arbitrage-driven fee income compensates an LP for the structural cost of AMM rebalancing. The answer, in a transparent Python framework, is that compensation holds in three regimes: low volatility with high fees, mean-reverting prices, and pools with substantial non-arbitrage volume. Outside these regimes, including the illustrative ETH/USDC-style baseline at a 0.30% fee, the model's fee income does not fully compensate for the structural loss.

The regime map this project produces reproduces, in stylised form, the design logic of Uniswap V3's fee tiers, while the LVR analysis provides an academically sharper measure of the loss that fees are meant to compensate. The mismatch between fee income (linear in volatility) and the LVR cost (quadratic in volatility) emerges as the fundamental analytical obstacle to LP profitability in volatile pairs, and the broader observation that price-process structure matters as much as volatility itself is, in this author's view, the most useful single takeaway for the design and selection of AMM liquidity positions in practice.

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